



ORIGINAL RESEARCH ARTICLE

Remote Facilitation of Essential Newborn Care: A Multinational, Multicenter Pilot Study

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ABSTRACT

BACKGROUND AND OBJECTIVES: Essential Newborn Care (ENC) training improves neonatal outcomes, particularly in low-resource settings. The American Academy of Pediatrics and Laerdal Global Health developed an online platform, <https://hmbs.org/>, to facilitate training of health care workers (HCWs) in ENC. This study aims to examine the impact of training with the ENC course. We hypothesize that training with the ENC course, either with remote or in-person facilitation, and low-dose high-frequency (LDHF) practice improves HCW knowledge and skills.

METHODS: In this prospective educational intervention study, technical advisors remotely oriented in-country facilitators to ENC in Nigeria and Bangladesh. In-country facilitators then trained frontline HCWs, choosing between a remote or in-person approach based on the country context. ENC knowledge check, bag-mask ventilation (BMV) skills, and NeoNatalie Live (NNL) manikin feedback pass rates were assessed at baseline (BL), immediately posttraining (PT), and endline (EL). Objective structured clinical examination (OSCE) A and B scores were assessed at PT and EL. LDHF practice was implemented at all sites using the NNL manikin.

RESULTS: After remote orientation, 18 in-country facilitators trained 236 frontline HCWs at 8 sites, including 1 humanitarian setting. All sites showed significant improvement in pass rates from BL to PT in knowledge check and BMV skills. NNL pass rates generally improved from PT to EL. OSCE A and B pass rates were also generally high PT and maintained at EL.

CONCLUSIONS: ENC online materials coupled with LDHF practice can augment knowledge and skills in ENC and offer a flexible option for remote or in-person training.



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Niermeyer and Savich participated in the project as technical advisors, shaped the content of the manuscript, analyzed data, and edited the manuscript. Mr Hattar served as the primary American Academy of Pediatrics project manager for this project and edited the manuscript.

(Continued)

WHAT'S KNOWN ON THIS SUBJECT: Essential Newborn Care training improves neonatal outcomes, particularly in low-resource settings. The American Academy of Pediatrics and Laerdal Global Health developed an online platform, hmbs.org, to facilitate remote training of health care workers.

WHAT THIS STUDY ADDS: This study examines the impact of an online platform coupled with low-dose high-frequency practice on health care workers' skills and knowledge in Essential Newborn Care.

Introduction

In 2000, the United Nations (UN) Millennium Development Goals catalyzed improved child and newborn health, leading to decreased under-5 years old mortality worldwide.^{1,2} The Every Newborn Action Plan and subsequent UN Sustainable Development Goals call for ending preventable newborn deaths, with all countries aiming to reduce neonatal mortality to 12 per 1000 live births or below by 2030.¹ Despite this, in 2022, sub-Saharan Africa and Southern Asia still had the highest neonatal mortality rates globally, with 27 deaths per 1000 live births in sub-Saharan Africa and 21 deaths per 1000 live births in Southern Asia.³⁻⁵

In the early 2010s, the American Academy of Pediatrics (AAP), in partnership with Laerdal Global Health (LGH), developed Helping Babies Survive (HBS), designed to improve knowledge and skills of birth attendants in resource-limited settings and support retention of lifesaving skills through low-dose high-frequency (LDHF) practice sessions. HBS courses have been disseminated in over 80 countries.⁶⁻¹³ Several recent systematic reviews show that standard formal neonatal resuscitation training improves outcomes, decreasing intrapartum-related deaths and early neonatal mortality. Additionally, multiple individual studies and meta-analyses demonstrate that the stillbirth rate, very early neonatal mortality, and overall neonatal mortality improved after Helping Babies Breathe (HBB) training.¹²⁻¹⁹ Thus, training is a vital tool that may help countries reach the UN Sustainable Development Goals.

In 2020, the COVID-19 pandemic halted in-person education. In response, the AAP and LGH developed an online platform (Helping Mothers and Babies Survive [HMBS] at <https://hmbs.org/>) to support remote facilitation of the Essential Newborn Care (ENC) Course 2nd edition, allowing for continued training of those involved in newborn resuscitation and care. Both AAP and

LGH collaborated with the World Health Organization (WHO) to transition the HBS learning methodologies into the ENC course and obtained WHO approval to make the draft content, ENC Now!, available during the pandemic. The objective of this study was to determine if the ENC online platform, coupled with LDHF practice, was effective in improving and maintaining health care workers' (HCWs') knowledge and skills across multiple sites in Nigeria and Bangladesh.

Methods

This prospective educational intervention study evaluated HCW knowledge and skills before and after ENC training (either remote or in person) in 2 countries: Nigeria and Bangladesh. Nigeria and Bangladesh were selected as priority countries with high neonatal mortality rates and established relationships between the national pediatric societies and the AAP. There was intentional inclusion of sites with conflict in northern Nigeria and deliberate choice of a humanitarian setting in Bangladesh.

Three distinct groups participated: (1) 4 sites in Nigeria (Borno State Specialist Hospital in Maiduguri, Yobe State Specialist Hospital in Damaturu, Gombe State Specialist Hospital in Gombe, and Yobe-Federal Medical Center in Nguru); (2) 3 sites in Dhaka, Bangladesh (Madaripur District Hospital, Manikganj District Hospital, and Mohammadpur Fertility Services and Treatment Center); and (3) 13 health clinics in the refugee camps in Cox's Bazar, Bangladesh (Table 1). All Nigerian sites are secondary referral centers. Mohammadpur Fertility Services and Treatment Center in Dhaka, Bangladesh, is a tertiary referral center, whereas the other 2 Dhaka sites are district hospitals. The Cox's Bazar sites are a mix of nongovernment organizations and government facilities providing services to 34 refugee camps located in the Ukhiya and Teknaf Upazilas.

TABLE 1. Characteristics of Sites and Participants

Location	Health Facility	Training Type	Health Setting	Facilitators	HCW Trained
Nigeria				3	122
	Yobe State Specialist Hospital Damaturu	Remote	Urban		31
	Federal Medical Center Nguru	Remote	Semi-urban		31
	Gombe State Specialist Hospital	Remote	Urban		30
	Borno State Specialist Hospital in Maiduguri	Remote	Urban		30
Dhaka, Bangladesh				12	86
	Mohammadpur Fertility Services and Treatment Center	Remote	Urban		56
	Manikganj District Hospital	Remote	Semi-urban		15
	Madaripur District Hospital	Remote	Semi-urban		15
Cox's Bazar, Bangladesh				3	28
	13 health clinics serving 34 refugee camps	In-person	Humanitarian		28

Abbreviation: HCW, health care worker.

Implementing partners assisted in choosing sites for inclusion, providing study oversight, and performing data collection. In Nigeria, the Paediatric Association of Nigeria and InSiGHT Health Consulting served as technical and implementing partners, respectively. Implementing partners in Bangladesh were Bangabandhu Sheikh Mukib Medical University and Save the Children Federation. In Cox's Bazar, MedGlobal and Hope Foundation for Women and Children of Bangladesh served as the implementing partners.

The AAP recruited 6 technical advisors (TAs) from 3 countries (Canada, Ethiopia, and the United States) to serve as remote facilitators. Selection was based on their prior experience as HBB facilitators in low- and middle-income countries (LMICs), career stage (late and early), and country of residence.

Development of Educational Materials

Remote ENC training sessions before this study used a slide presentation of the WHO and AAP ENC draft, which was large, noninteractive, and not easily downloadable. Development of the interactive online materials sought to preserve the active adult learning methodologies of HBS. Empowering learners with peer-to-peer learning, case-based scenarios, and repeated demonstration and practice were elements from HBB that were maintained for this ENC version. New elements of the ENC course included live birth and demonstration videos, engaging graphics, interactive quizzes, facility-based education, and emphasis on respectful care, infection control, and recording and using data.

LGH led the online platform creation, and AAP convened a small workgroup of experienced HBB and ENC facilitators to review, test, and provide feedback of the first part of ENC (Immediate Care and Helping Babies Breathe at Birth) to ensure usability, interactivity, and learner engagement. The core learning materials were

- Online ENC content: key content, actual birth and demonstration videos, and discussion questions
- Action plan: poster showing the steps to take during care (Supplemental Figure 1)
- Facilitator flip chart: facilitation tool illustrating concepts for participants and providing guidance for facilitators to demonstrate, lead discussions, and support practice (Supplemental Figure 2)
- Provider guide: booklet for every HCW to review content and practice skills during the training (Supplemental Figure 3)
- Supplemental learning materials: checklists, facilitation guides, simulation practice cards (Supplemental Figure 4), and assessment guides

Because of internet connectivity concerns, materials were downloadable for digital viewing offline. Supplies shipped to the sites included Wi-Fi hotspots, NeoNatalie Live (NNL) manikins (version 5.9.1, firmware version 1.0.0.4), resuscitation kits, and

tablets. The NNL is the sensorized version of the NeoNatalie simulator. Additional supplies for neonatal resuscitation were provided to Cox's Bazar, including stethoscopes, penguin suction devices, blankets, and caps.

Training Cascade

In remote facilitation, facilitators lead the ENC session through an online conferencing platform. The 6 TAs remotely oriented in-country facilitators to ENC using Zoom and WhatsApp Chat. In-country facilitators all had previous HBB experience. The orientation included introduction to the learning materials, preparation for birth, routine care at birth, the golden minute, continued ventilation, and discussion on sustainability through quality improvement practices.

The platform and online materials allowed continued training via remote facilitation, but they also could be used for in-person instruction. The in-country facilitators cascaded remote or in-person trainings to frontline HCWs. In Nigeria and Dhaka, Bangladesh, distance and travel restrictions during COVID-19 warranted remote facilitation. In Cox's Bazar, limited network connectivity in the camps necessitated in-person training.

Learners, selected by the in-country facilitators and implementing partners, were direct frontline HCWs at the health facilities, often providing newborn resuscitation and care as part of their duties. Each site customized their training to occur over several days to weeks. HCWs initiated LDHF practice sessions, depending on the site. LDHF practice promotes retention of clinical knowledge and skills through multiple short, targeted, in-service simulation-based activities. When done in groups or pairs, peer-to-peer feedback is vital for LDHF practice. Sessions lasted less than 15 minutes, focused on critical skills of neonatal resuscitation, and varied from daily to weekly to twice a month.

Assessments

There were 4 educational assessments: a knowledge check, a bag-mask ventilation (BMV) skills check, objective structured clinical examination (OSCE) A and B, and feedback from the NNL manikin. Assessments occurred at 3 time points: at baseline (BL) before training began, immediately posttraining (PT), and at endline (EL) up to 6 months after the end of the initial training. Between PT and EL, participants practiced in LDHF sessions.

With the NNL, participants performed BMV and received an indication of successful resuscitation, passing the assessment, or constructive feedback to improve resuscitation and failing the assessment. Specific feedback included the rate of ventilation, tidal volume, mask leak, and any pauses or interruptions. The Liveborn tablet-based software application provided feedback on ventilation performance. NNL was used to assess participants during the study, as well as during LDHF practice.

The knowledge check, BMV skills, and NNL assessment were performed at BL, PT, and EL, and the OSCEs were performed at

PT and EL to determine retention of knowledge and skills. The HMBS assessment platform is a web-based application for assessment and recording of scores. Assessment details are available in Supplemental Materials (Supplemental 1).

Statistical Analysis

The percentage of participants passing each test at each time point was compared. Cochran's Q test was performed for the outcome (pass or fail) across all 3 time points, and McNemar test was performed for the outcome (pass or fail) across 2 given time points.²⁰ Certain items on the OSCEs must be answered correctly to pass regardless of overall score, and NNL feedback is classified as pass or fail. For the knowledge check, critical concepts exist for lifesaving knowledge; if learners did not answer certain questions correctly despite answering other areas correctly, they failed. For the BMV skills, learners needed to complete all skills to pass.

Given site-specific implementation differences, data were analyzed per site. In Cox's Bazar, all health clinics serve the same humanitarian setting; thus, data were aggregated as 1 site. For the knowledge check and BMV checklist, the percentage of participants passing was compared across time points using Cochran's Q test. Any site with a significant Cochran's Q test difference was tested using the pairwise McNemar test for each pair of time points. To correct for multiple time points, the Bonferroni correction was applied to the significance level (alpha) for the McNemar test. The OSCEs were assessed at

2 time points, so only the pairwise McNemar test was needed. NNL data were not identified; aggregate, descriptive results for each site are presented. Analysis was conducted using IBM SPSS Statistics for Windows (version 29.0.2.0).

Ethical Considerations

The AAP Institutional Review Board, Bangabandhu Sheikh Mujib Medical University Institutional Review Board in Bangladesh, and the National Health and Ethics Research Committee in Nigeria approved this study.

Results

TA facilitation of ENC for in-country facilitators in Nigeria (n = 3), Dhaka, Bangladesh (n = 12) and Cox's Bazar, Bangladesh (n = 3), occurred in late 2021. In-country facilitators then cascaded training at their sites, for a total of 236 HCWs trained (Table 1). Figure 1 provides an overview of the study timeline. Of the 122 HCWs at the Nigeria sites, there were 62 registered midwives (51%), 33 physicians (27%), 16 registered nurses (13%), 6 pediatric nurses (5%), 4 anesthesia nurses (3%), and 1 critical care nurse (1%). Of the 86 HCWs at the Dhaka, Bangladesh sites, there were 42 registered nurses (49%), 27 physicians (31%), 9 family welfare visitors (10%), and 8 registered midwives (9%). Out of the 28 HCWs trained at the Cox's Bazar, Bangladesh sites, all were registered midwives (100%).

Educational outcomes for the knowledge test, BMV skills, NNL scenario, and OSCE A and B are reported in Figure 2A to 2E.

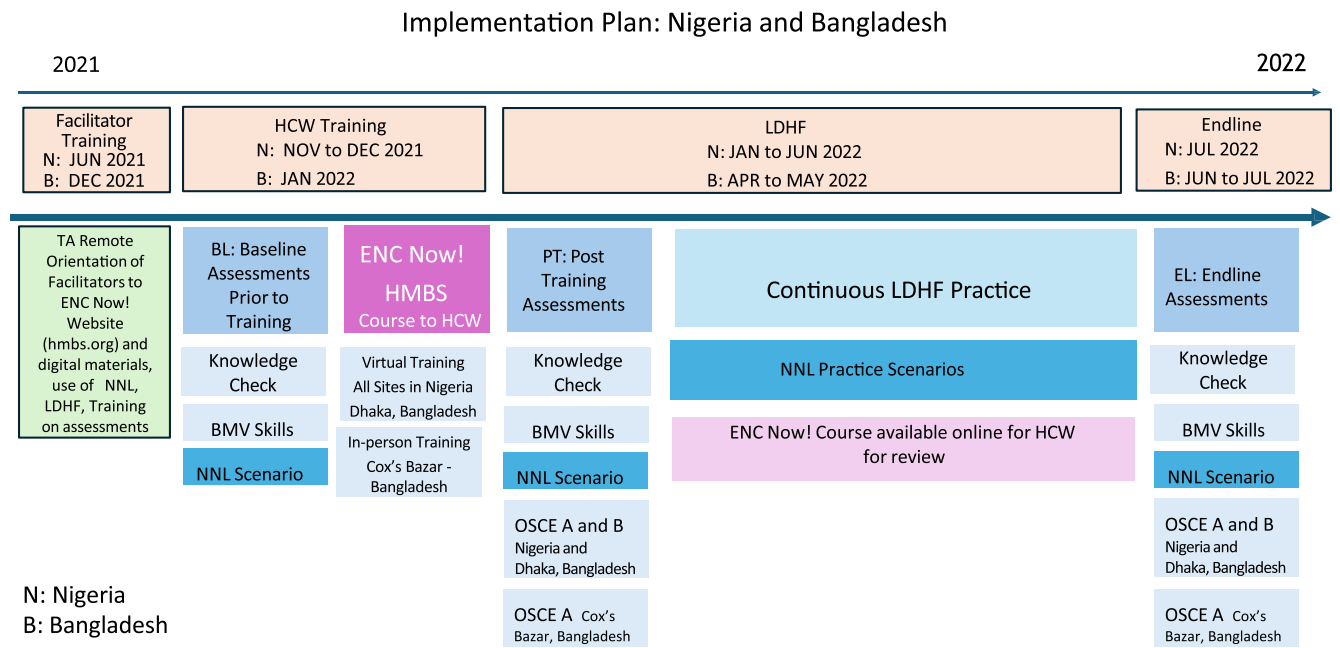


FIGURE 1. Study timeline: implementation plan and timeline for each study phase. Abbreviations: BL, baseline; BMV, bag-mask ventilation; EL, endline; ENC, Essential Newborn Care; HCW, health care workers; HMBS, Helping Mothers and Babies Survive; LDHF, low-dose high-frequency; NNL, NeoNatalie Live; OSCE, objective structured clinical examination; PT, posttraining; TA, technical advisor.

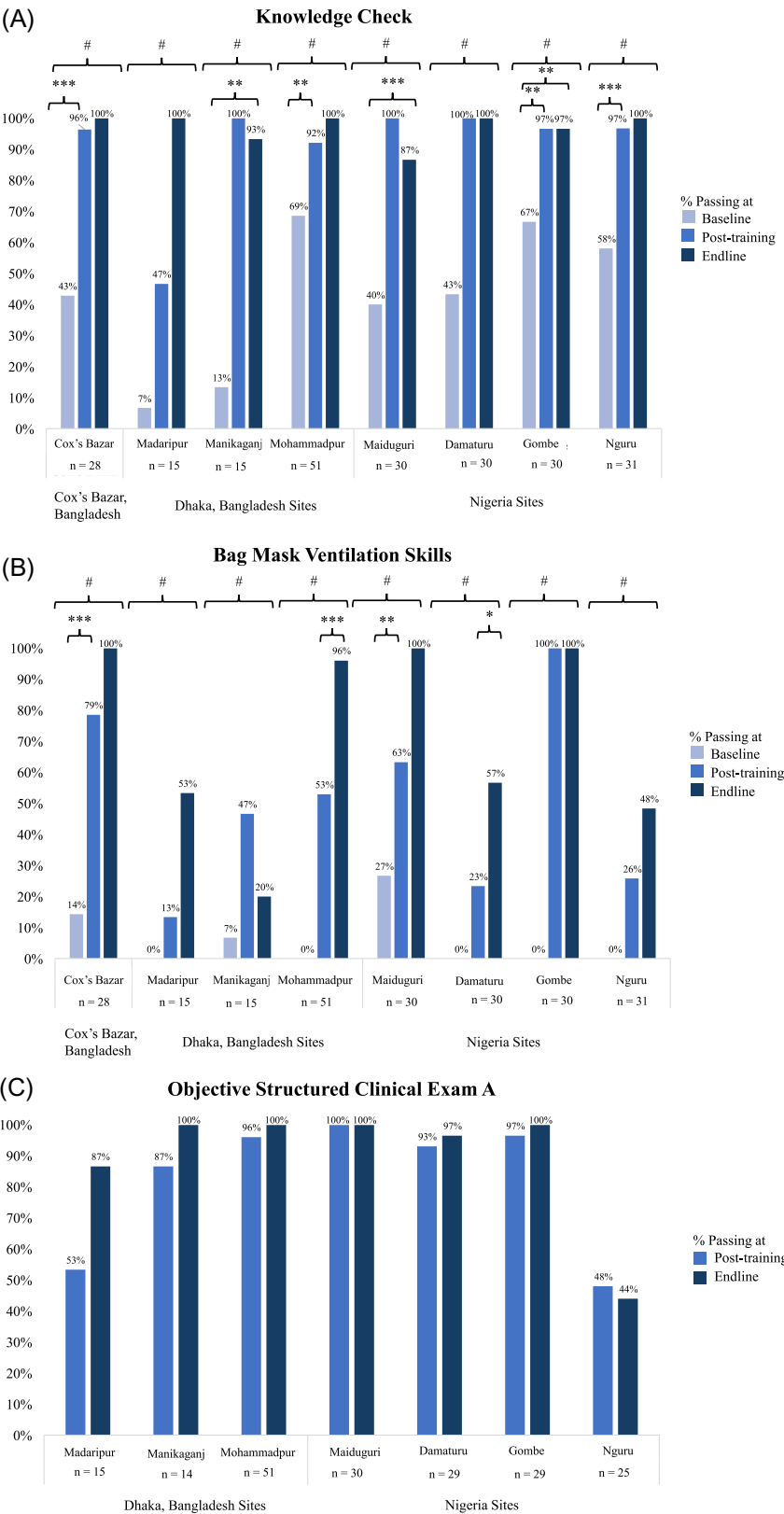


FIGURE 2. (Continued).

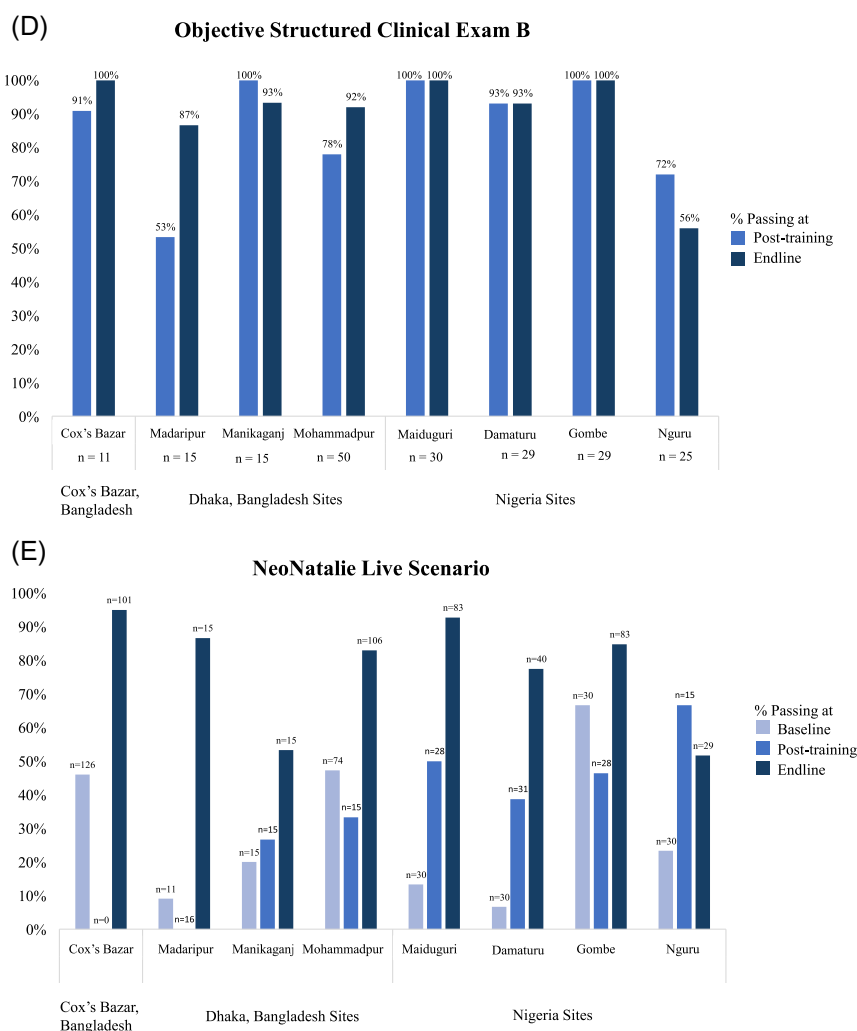


FIGURE 2. Educational assessments pass rates: health care workers took the tests before training (baseline), immediately posttraining, and at endline. Objective structured clinical examination (OSCE) A and B were not tested at baseline. OSCE A was not tested at Cox's Bazar. Pass rates are shown with significance levels from the Cochrane's Q test analysis and the McNemar test with Bonferroni correction. Note: the McNemar test cannot calculate significance when 1 time point has no variation (eg, all individuals passed the test or all failed). Where all individuals passed the test and significance is not shown, the result may not be insignificant, but significance cannot be calculated. (A) Knowledge check pass rates by site. (B) Bag-mask ventilation skills pass rate by site. (C) OSCE A pass rate by site. (D) OSCE B pass rate by site. (E) NeoNatalie live scenario pass rate by site. McNemar test, * $P < .0167$, ** $P < .01$, *** $P < .001$; Cochrane's Q test analysis, # $P < .05$.

All sites showed significant improvement on the knowledge check and BMV checklist across all time points. It is important to note that the McNemar test cannot be performed where there is no variation, which was frequently the case in which either all participants failed at BL or all passed at PT or EL. In these cases (for example, at Damaturu in Nigeria where 43% passed the knowledge check at BL and 100% passed at PT and EL), although significance cannot be reported, the size of the change suggests that the result is not insignificant.

For OSCE A, immediately PT, the pass rates ranged from 48% to 100%. These pass rates were maintained at EL and ranged from 44% to 100% (1 site declined at EL). Similar findings were

seen for OSCE B and ranged from 53% to 100% at PT and 56% to 100% at EL. At all sites, the OSCE showed no statistically significant improvement from PT to EL (some sites had no variation so were not subject to the McNemar test). For the NNL scenario, all but 1 site (Nguru) saw a large improvement in the average ability to resuscitate the NNL from BL to EL (median improvement 42%). There was less improvement between BL to PT (median improvement 7%) than from PT to EL (median improvement 39%). Some participants practiced with the NNL scenario multiple times at each time point, resulting in more recorded sessions compared with the numbers of actual HCW trained at each site.

HCWs performed LDHF practice with the NNL manikin from the PT to the EL period. These sessions were conducted by all centers but varied widely in average session per learner (in Cox's Bazar, 8.2; in Dhaka, 6.8; and in Nigeria, 68.3). NNL manikins were always available for LDHF practice at all Nigeria sites in the labor wards during dayshift hours. In Dhaka, Bangladesh, NNL manikins were available for practice in the labor ward during multiple scheduled practice times each week and, when requested, outside the scheduled times. In Cox's Bazar, Bangladesh, the NNL manikin was available for practice in the labor ward every day during dayshift hours.

Discussion

Many years of experience have informed the strengths and challenges of newborn care education in LMICs.^{6,21} The COVID-19 pandemic and resultant travel and meeting constraints led to the development of an online platform and digital materials to enable remote skills training. This pilot study of the feasibility of remote or in-person facilitation of the online ENC course demonstrates improvements in educational outcomes across multiple sites for knowledge assessment, BMV skills, and OSCE and NNL assessments.

There were significant improvements in the knowledge check and BMV skills between BL and PT, which were maintained at the EL testing. OSCE assessment scores were generally high PT and maintained at EL. Passing scores with the NNL assessment were generally much higher comparing EL with PT, likely because the LDHF practice between these 2 time points also used the NNL manikin. The role of LDHF practice in sustaining skills has been well demonstrated in the literature. A recent study on LDHF practice for neonatal resuscitation in a high-income country showed that increased training frequency was important in maintaining neonatal ventilation competence.²² In Uganda, peer-assisted LDHF practice resulted in significant skills maintenance compared with control for postpartum hemorrhage and birth asphyxia prevention in 125 facilities.²³ Another study based in India showed improved knowledge retention with LDHF practice compared with control for initial steps of newborn resuscitation.²⁴ A smaller study from Mozambique found that 5 months of LDHF practice improved clinical care during newborn resuscitation.²⁵ This pilot study of ENC mirrors the findings from previous studies of LDHF practice with preservation of skills and knowledge for up to 6 months, contrary to available evidence that trained HCWs often experience a decay of skills and knowledge after training.^{26–28}

Multiple studies, many of which were conducted during the COVID-19 pandemic, have shown the effectiveness of remote learning.^{29–32} Previous studies involving fewer than 30 learners examined the impact of remote facilitation for resuscitation training both in high-income countries and LMICs, showing that it is feasible and may be cost-effective.^{33–36} This remote learning methodology augments the data from several systematic

analyses that demonstrate in-person training in neonatal resuscitation and HBB impacts on perinatal outcomes, including decreased intrapartum death, stillbirth rate, early neonatal mortality, and overall neonatal mortality.^{12–19,37}

Strengths of the study included multicountry, multisite inclusion, which shows that the online materials are adaptable and allowed for flexibility, especially in traditionally hard-to-access humanitarian settings like refugee camps. Implementing partners and in-country facilitators were instrumental in helping organize educational sessions, check equipment, and collect educational data. This diverse partnership network enhanced the study's capacity to adapt and continue in challenging field conditions.

This study faced limitations. The study logistics were affected by uncontrollable factors during the pandemic such as supply chain issues, clinical workload of learners, multiple government-mandated COVID-19 lockdowns, and worker strikes. These issues delayed the study timeline, may have led to staff turnover, and limited the period for LDHF practice in Bangladesh. For future studies of ENC, careful attention to timely data collection is strongly recommended, with particular attention to timely assessment, avoiding long lag times between study points, and tracking of staff turnover, which limits paired data.

Further limitations on this study's reproducibility include lack of details regarding HCWs' experience, their specific frontline responsibilities, previous training in neonatal resuscitation or HBB, and details about the length of each subsequent ENC training. Data on LDHF practice, including how many sessions were initiated by HCWs vs facilitators, were also not recorded. Sites in Nigeria had a longer period between PT and EL assessments compared with Bangladesh because of a variety of factors beyond the control of study-implementing partners and thus recorded higher numbers of NNL manikin practice sessions compared with all sites in Bangladesh.

Despite the intention for remote facilitation, in Cox's Bazar camps, there was limited Wi-Fi availability. The Cox's Bazar implementers' use of the materials in face-to-face education demonstrates the adaptability of the ENC platform to a variety of facilitation methods. Although this study provided proof of concept of the use of online facilitation to better reach humanitarian settings, for future remote work, funding for reliable Wi-Fi at sites is another strong recommendation.

In this real-world setting, the study showed feasibility of remote and in-person training in LMIC and hard-to-access sites using an online ENC platform for customized, freely accessible learning materials. Overall, this study demonstrated promising outcomes regarding the efficacy of remote and in-person facilitation of the online ENC curriculum. Future studies may address the clinical impact of remote facilitation, such as reductions in neonatal mortality, birth asphyxia, and stillbirth rates, which were beyond the scope of this study. In conclusion,

this study showed facilitation of ENC using an online platform coupled with LDHF practice improves knowledge and skills in HCWs. Digital materials and remote facilitation can revitalize professional education in ENC under pandemic constraints, especially in hard-to-access areas.

Abbreviations

AAP: American Academy of Pediatrics
 BL: baseline
 BMV: bag-mask ventilation
 EL: endline
 ENC: Essential Newborn Care
 HBB: Helping Babies Breathe
 HBS: Helping Babies Survive
 HCW: health care worker
 HMBS: Helping Mothers and Babies Survive
 LDHF: low-dose high-frequency
 LGH: Laerdal Global Health
 LMIC: low- and middle-income country

NNL: NeoNatalie Live
 OSCE: objective structured clinical examination
 PT: posttraining
 TA: technical advisor
 UN: United Nations
 WHO: World Health Organization

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Ms Littman served as the statistician for this project, analyzed the data, shaped the content of the manuscript, and revised the manuscript. Dr Alex-Adeomi provided managerial oversight for this project, shaped the content of the manuscript, and revised the manuscript. Dr Patterson served as senior American Academy of Pediatrics leadership, providing strategic guidance on project conception and implementation. Dr Mannan, Dr Chowdhury, Dr Dey, Dr Jahan, Ms Khatun, Drs Moni, and Hassan Shabuj served as the team of in-country facilitators for the 3 sites in Dhaka, Bangladesh. Dr Shahidullah provided leadership for this team. Dr Oishee served as the primary research coordinator for Save the Children, the in-country implementation team for the 3 sites in Dhaka, Bangladesh. Mr Akala, Dr Mba-Oduwusi, and Mr Rahmon served as in-country implementation team in Nigeria and provided project management and quality assurance. Drs Farouk, Garba, and Raymond Poksireni served as the team of in-country facilitators in Nigeria. Dr Tabansi provided leadership for this team. Dr Hafiz, Dr Hoque, and Mr Zahiduzzaman conducted data collection for the Cox's Bazar sites. Dr Mahmood provided leadership for this team. Ms Khatun and Ms Nurunnaher served as the team of in-country facilitators for Cox's Bazar sites. Dr Molla provided leadership and technical support to this team. All authors approved the final manuscript as submitted and agree to be accountable for all aspects of the work.

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